

Air University



Cryptography (Lab)

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Lab # 09

Comparative Study of AES and DES Encryption Algorithms

1. Theory

AES (Advanced Encryption Standard)

AES, or Advanced Encryption Standard, is a symmetric key block cipher recognized by NIST. It processes data in fixed-size 128-bit blocks and supports key lengths of 128, 192, or 256 bits. Rather than using a Feistel structure, AES relies on a Substitution-Permutation Network (SPN) to transform plaintext into ciphertext through multiple rounds of substitution, permutation, and mixing. The number of rounds varies by key size:

- 10 rounds for AES-128
- 12 rounds for AES-192
- 14 rounds for AES-256

The popularity of AES in today's systems is due to its:

- Strong security
- High efficiency in both hardware and software
- Resilience against known attacks

DES (Data Encryption Standard)

The Data Encryption Standard (DES) is an older symmetric block cipher that operates on 64-bit blocks and utilizes a 56-bit key. DES is structured as a Feistel Network, dividing plaintext into two halves and processing them over 16 rounds using substitution and permutation techniques. Although DES was once the gold standard, it is now deemed insecure because of:

- A key size that is too small, making it vulnerable to brute-force attacks
- The rapid increase in computing power

To address these security concerns, Triple DES (3DES) was developed, which applies DES three times with different keys to bolster encryption strength.

2. Tools & Requirements

Programming Language : None

IDE : Online Tools

Links : AES <https://legacy.cryptool.org/en/cto/aes-step-by-step>

DES <https://simewu.com/des/>

3. AES Algorithm Steps

i. Key Expansion

Expanded Key

```
00000000 00000000 00000000 00000000 62636363 62636363 62636363 62636363 9b9898c9 f9fbfbba 9b9898c9 f9fbfbba 90973450
696ccffa f2f45733 0b0fac99 ee06da7b 876a1581 759e42b2 7e91ee2b 7f2e2b88 f8443e09 8dda7cbb f34b9290 ec614b85 1425758c
99ff0937 6ab49ba7 21751787 3550620b acaf6b3c c61bf09b 0ef90333 3ba96138 97060a04 511dfa9f b1d4d8e2 8a7db9da 1d7bb3de
4c664941 b4ef5bcb 3e92e211 23e951cf 6f8f188e
```

ii. Initial Round

Round 1

input to Round 1			
5573616d	61000000	00000000	00000000
after S-Box:			
fc8fef3c	ef636363	63636363	63636363
after permutation:			
fc636363	ef63633c	6363ef63	638f6363
after mult:			
46fcfcd9	3fb00e52	efec60ef	4ca08f8f
used subkey:			
62636363	62636363	62636363	62636363
after mix with key:			
249f9fba	5dd36d31	8d8f038c	2ec3ecec

iii. Multiple Rounds

Round 2

input to Round 2			
249f9fba	5dd36d31	8d8f038c	2ec3ecec
after S-Box:			
36dbdbf4	4c663cc7	5d737b64	312ecece
after permutation:			
36667bce	4c73cef4	5d2edbc7	31db3c64
after mult:			
73b9efc0	3717bf9a	d4b08c87	4cbc3e7c
used subkey:			
9b9898c9	f9fbfbba	9b9898c9	f9fbfbba
after mix with key:			
e8217709	ceec4430	4f28144e	b547c5d6

Round 3

input to Round 3

e8217709 ceec4430 4f28144e b547c5d6

after S-Box:

ON

9bdfd501 8bce1b04 8434fa2f d5a0a6f6

after permutation:

ON

9bcefaf6 8b34a601 84a0f504 d5fd1b2f

after mult:

ON

68ffbb75 f613eb16 19dfd9ca 99366fdc

used subkey:

90973450 696ccffa f2f45733 0b0fac99

after mix with key:

ON

f8688f25 9f7f24ec eb2b8ef9 9239c345

iv. Final Round

Round 10

input to Round 10

257bf0f6 928c1010 ad84445a 6faac26c

after S-Box:

ON

3f218c42 4f64caca 955f1bbe a8ac2550

after permutation:

ON

3f641b50 4f5f2542 95ac8cca a821cabe

used subkey:

b4ef5bcb 3e92e211 23e951cf 6f8f188e

after mix with key:

ON

8b8b409b 71cdc753 b645dd05 c7aed230

4. DES Algorithm Steps

1. Initial Permutation

Shuffle bits according to the initial permutation (IP):

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

Original message (64-bit):

0101 0101 0111 0011 0110 0001 0110 1101 0110 0001 0000 0000 0000 0000 0000 0000

Message after IP (64-bit):

0001 1111 0000 0011 0000 1001 0001 1111 0000 0000 0001 1110 0000 1000 0000 0010

2. 16 Feistel Rounds

Step 8: Computing ROUND 1

$L_0 = 0001\ 1111\ 0000\ 0011\ 0000\ 1001\ 0001\ 1111$

$R_0 = 0000\ 0000\ 0001\ 1110\ 0000\ 1000\ 0000\ 0010$

$K_1 = 000010\ 110000\ 001001\ 100111\ 100110\ 110100\ 100110\ 100101$

$E(R_0) = 000000\ 000000\ 000011\ 111100\ 000001\ 010000\ 000000\ 000100$

$K_1 \oplus E(R_0) = 000010\ 110000\ 001010\ 011011\ 100111\ 100100\ 100110\ 100001$

$S(K_1 \oplus E(R_0)) = 0100\ 0101\ 0011\ 1010\ 0111\ 1111\ 1101\ 0010$

$f = P(S(K_1 \oplus E(R_0))) = 0011\ 0110\ 0111\ 0110\ 1110\ 0000\ 1101\ 1101$

$L_1 = R_0 = 0000\ 0000\ 0001\ 1110\ 0000\ 1000\ 0000\ 0010$

$R_1 = L_0 \oplus f(R_0, K_1) = 0010\ 1001\ 0111\ 0101\ 1110\ 1001\ 1100\ 0010$

Step 9: Computing ROUND 2

$$L_1 = 0000\ 0000\ 0001\ 1110\ 0000\ 1000\ 0000\ 0010$$

$$R_1 = 0010\ 1001\ 0111\ 0101\ 1110\ 1001\ 1100\ 0010$$

$$K_2 = 011010\ 011010\ 011001\ 011001\ 001001\ 010110\ 101000\ 100110$$

$$E(R_1) = 000101\ 010010\ 101110\ 101011\ 111101\ 010011\ 111000\ 000100$$

$$K_2 \oplus E(R_1) = 011111\ 001000\ 110111\ 110010\ 110100\ 000101\ 010000\ 100010$$

$$S(K_2 \oplus E(R_1)) = 1000\ 0110\ 0011\ 0001\ 1100\ 0100\ 0011\ 1011$$

$$f = P(S(K_2 \oplus E(R_1))) = 1100\ 1111\ 1000\ 0110\ 0000\ 1100\ 0001\ 1100$$

$$L_2 = R_1 = 0010\ 1001\ 0111\ 0101\ 1110\ 1001\ 1100\ 0010$$

$$R_2 = L_1 \oplus f(R_1, K_2) = 1100\ 1111\ 1001\ 1000\ 0000\ 0100\ 0001\ 1110$$

3. Expansion Function

Step 7: Encode the data

Splitting the resulting message into two halves:

$$L_0 = 0001\ 1111\ 0000\ 0011\ 0000\ 1001\ 0001\ 1111$$

$$R_0 = 0000\ 0000\ 0001\ 1110\ 0000\ 1000\ 0000\ 0010$$

This can be expanded to 16 entries using the following formula

$$L_n = R_{n-1}$$

$$R_n = L_{n-1} \oplus f(R_{n-1}, K_n)$$

Where $f = P(S(K_n \oplus E(R_n)))$, where E expands 32 bits into 48 bits using the following table:

32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

4. Substitution (S-box)

Then S splits the 48 bits into eight groups of six bits, and each group gets ran through the S-box as follows:

Given the first group of six bits, $abcdef$, consider af to be the row number, and $bcde$ to be the column number in the following table:

S_1 :

14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

The other seven groups have the following S-boxes:

S_2 :

15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9

S_7 :

4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12

S_8 :

13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

5. Permutation

Then finally, P mixes up the 32-bit output to get the output of f , using the following table:

16	7	20	21
29	12	28	17
1	15	23	26
5	18	31	10
2	8	24	14
32	27	3	9
19	13	30	6
22	11	4	25

Using the formula described previously:

$$L_n = R_{n-1}$$

$$R_n = L_{n-1} \oplus f(R_{n-1}, K_n) \text{ where } f = S(K_n \oplus E(R_n)),$$

We apply them iteratively in a series of rounds.

6. Final Permutation

Step 24: Permute the encoded data through the IP^{-1} table

Now that we have

$$L_{16} = R_{15} = 0001\ 1100\ 0000\ 1001\ 0011\ 0000\ 1111\ 1101$$

$$R_{16} = L_{15} \oplus f(R_{15}, K_{16}) = 0110\ 1101\ 0111\ 1011\ 1101\ 1000\ 1110\ 0010$$

Concatenate them backwards to get:

$$R_{16}L_{16} = 0110\ 1101\ 0111\ 1011\ 1101\ 1000\ 1110\ 0010\ 0001\ 1100\ 0000\ 1001\ 0011\ 0000\ 1111\ 1101$$

Then re-shuffle $R_{16}L_{16}$ using the following IP^{-1} table:

40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

5. Result

AES

Input: plaintext “Usama”

```
Input
5573616d 61000000 00000000 00000000
```

Ciphertext

```
Encoded
8b8b409b 71cdc753 b645dd05 c7aed230
```

DES

Input: plaintext “Usama”

```
Message 5573616D61000000
```

Ciphertext

```
Ciphertext 7211C2F69E5B5707
```

Parameter	AES	DES
Key Size	128, 192, or 256 bits	56 bits (64 bits including parity)
Block Size	128 bits	64 bits
Encryption Time	Faster (efficient for both hardware and software)	Slower (less efficient, particularly in software)
Decryption Time	Fast (uses the same structure as encryption)	Slower (similar complexity to encryption)
Security Level	Very High (secure against brute-force attempts)	Low (vulnerable to brute-force attacks)

6. Conclusion

Security Comparison:

The Advanced Encryption Standard (AES) employs a 128-bit key, offering a staggering 2^{128} possible combinations, which renders brute-force attacks practically impossible. In contrast, the Data Encryption Standard (DES) relies on a 56-bit key, resulting in roughly 72 quadrillion (2^{56}) combinations. This key length was effectively compromised in 1998, with specialized hardware cracking it in under 24 hours. Today, AES is the accepted standard endorsed by NIST, while DES is fully retired from active use.

Speed Comparison:

Both algorithms are capable of encrypting data in microseconds for shorter messages. DES operates through 16 Feistel rounds on 8-byte blocks, whereas AES executes 10 rounds on 16-byte blocks. However, on modern

hardware, AES often outperforms DES thanks to AES-NI instructions that boost its speed. In the absence of these specialized instructions, both algorithms show similar performance when processed in software.

Practical Usage:

AES enjoys widespread implementation across numerous applications, including HTTPS/TLS, disk encryption solutions like BitLocker and FileVault, Wi-Fi security (WPA2/WPA3), VPNs, and government communications. On the other hand, DES has fallen out of favor in contemporary contexts and primarily lingers in legacy banking systems, where it has largely been supplanted by Triple DES (3DES). Current protocols necessitate the use of AES for new systems.

- End of Lab Task -